

Supervised Binary Classification for Candidate Job Selection via Regularized Logistic Regression

Machine Learning Foundations

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Bachelor's in Interdisciplinary Engineering

August 10, 2026

Abstract

Predicting job applicant hiring decisions based on quantitative performance metrics is a fundamental classification problem in automated recruitment systems. In this work, we present a lightweight binary logistic regression model trained from scratch using gradient descent with L_2 weight regularization. The model evaluates job applicants using two primary attributes: Academic Performance (CGPA) and Work Experience (years). We formulate the optimization objective using Binary Cross-Entropy (BCE) loss and derive the exact parameter updates. Using an 80/20 train-test split on a dataset of 300 candidates, the model achieves robust convergence within 20 epochs. Furthermore, we decouple the training pipeline from the inference engine by extracting the optimal weight vector w^* , enabling zero-dependency runtime predictions.

1 Introduction

Supervised machine learning algorithms provide automated mechanisms for decision-making by fitting parametric models to historical structured data. For binary classification tasks—such as determining whether a job candidate receives an offer ($y = 1$) or rejection ($y = 0$)—logistic regression serves as a foundational baseline due to its probabilistic interpretability and convex optimization properties.

In this project, we implement a linear binary classifier trained via batch gradient descent from first principles. By mapping raw continuous features through a non-linear activation function, the network learns a linear decision boundary in feature space.

1.1 Key Highlights

- **Custom ML Pipeline:** Implementation of logistic regression, loss evaluation, and parameter optimization from scratch using vectorised linear algebra.
- **Regularized Objective:** Incorporation of L_2 weight decay to prevent model overfitting while preserving bias term flexibility.
- **Model Decoupling for Deployment:** Separation of computationally heavier training operations from a minimal, closed-form inference function requiring zero external libraries.

2 Mathematical Formulation

We model candidate hiring probability as a generalized linear model parameterized by weights $\mathbf{w} \in \mathbb{R}^3$.

2.1 Feature Vector and Activation

Let each input candidate vector be defined as $\mathbf{x}^{(i)} = [1, x_1^{(i)}, x_2^{(i)}]^T$, where 1 is the bias offset, x_1 represents candidate CGPA, and x_2 represents total work experience in years.

The linear combination $z^{(i)}$ is transformed into a continuous probability prediction $p^{(i)} \in (0, 1)$ via the logistic sigmoid function $\sigma(z)$:

$$z^{(i)} = \mathbf{w}^T \mathbf{x}^{(i)} = w_0 + w_1 x_1^{(i)} + w_2 x_2^{(i)} \quad (1)$$

$$p^{(i)} = \sigma(z^{(i)}) = \frac{1}{1 + e^{-z^{(i)}}} \quad (2)$$

2.2 Loss Function with L_2 Regularization

To evaluate model performance, we utilize the Binary Cross-Entropy (BCE) loss function across N dataset samples, augmented by an L_2 weight regularization penalty parameter λ :

$$\mathcal{J}(\mathbf{w}) = -\frac{1}{N} \sum_{i=1}^N \left[y^{(i)} \ln(p^{(i)}) + (1 - y^{(i)}) \ln(1 - p^{(i)}) \right] + \frac{\lambda}{2} \sum_{j=1}^2 w_j^2 \quad (3)$$

Note that the bias parameter w_0 is explicitly excluded from the L_2 penalty to ensure unbiased baseline shifts.

2.3 Gradient Computation and Weight Updates

The gradient vector $\nabla_{\mathbf{w}} \mathcal{J}$ with respect to parameter vector $\mathbf{w}$ is calculated as:

$$\nabla_{\mathbf{w}} \mathcal{J} = \frac{1}{N} \mathbf{X}^T (\mathbf{p} - \mathbf{y}) + \lambda (\mathbf{w} \odot \mathbf{m}) \quad (4)$$

where $\mathbf{X} \in \mathbb{R}^{N \times 3}$ represents the design matrix, $\mathbf{m} = [0, 1, 1]^T$ masks the bias term from regularization, and $\odot$ denotes element-wise multiplication. Model parameters are updated iteratively given learning rate η :

$$\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} - \eta \nabla_{\mathbf{w}} \mathcal{J} \quad (5)$$

3 Implementation Details

3.1 Training Phase Configuration

The optimization procedure was executed over 20 epochs using matrix operations. Hyperparameters selected for optimal convergence are summarized in Table 1.

Table 1: Training Hyperparameters and Design Configurations

Parameter	Symbol	Value
Train / Test Split Ratio	—	80%/20%
Learning Rate	η	0.03
L_2 Regularization Factor	λ	0.01
Decision Threshold	τ	0.50
Total Training Epochs	E	20
Initial Weights Vector	$\mathbf{w}_{\text{init}}$	$[-8.0, 2.0, 0.6]^T$

3.2 Code Architecture

Figure 1 shows the deployment logic extracted for the inference script. The prediction phase bypasses external tensor/dataframe dependencies by relying directly on scalar math operations.

```

1 import math
2
3 def predict(cgpa, experience):
4     # Hardcoded optimal weights extracted post-training
5     bias = -8.14405058
6     w_cgpa = 1.14851008
7     w_exp = 0.29156678
8
9     z = bias + (w_cgpa * cgpa) + (w_exp * experience)
10    probability = 1.0 / (1.0 + math.exp(-z))
11    label = "YES" if probability >= 0.5 else "NO"
12
13    return probability, label

```

Figure 1: Isolated inference function utilizing exported model weights.

4 Results and Weight Convergence

Following 20 training epochs, the model converged to stable parameter values. The resulting learned parameter vector $\mathbf{w}^*$ is detailed in Table 2.

Table 2: Optimal Parameters Learned by Model

Parameter	Associated Feature	Learned Weight	Interpretation
w_0	Intercept / Bias	-8.1441	High baseline rejection threshold
w_1	Candidate CGPA (x_1)	+1.1485	Strong positive influence on hiring
w_2	Experience (x_2)	+0.2916	Moderate positive influence on hiring

4.1 Decision Boundary Analysis

Setting the classification decision rule to $p \geq 0.5$ corresponds to $z = 0$. Solving for the decision boundary in terms of feature x_2 (Experience) yields:

$$w_0 + w_1x_1 + w_2x_2 = 0 \implies x_2 = -\frac{w_1}{w_2}x_1 - \frac{w_0}{w_2} \quad (6)$$

Substituting the learned parameters yields the linear tradeoff equation:

$$\text{Experience} \approx -3.939 \cdot \text{CGPA} + 27.932 \quad (7)$$

This decision boundary illustrates that higher academic performance compensates for lower work experience when evaluating candidate eligibility.

5 Conclusion

This project demonstrates that a self-implemented logistic regression classifier using L_2 -regularized gradient descent effectively models applicant selection criteria. Extracting the parameter vector

$\mathbf{w}^*$ post-training allows for an ultra-lightweight deployment script that executes rapid inference without external library dependencies.